1. Purpose and Scope

The Riverbend Water Authority prepared this plan to guide management of the Riverbend Reservoir from 2026 through 2035. The reservoir has a total capacity of 48.5 million cubic meters.

The reservoir supplies drinking water to 310,000 residents in four municipalities: Alder, Brook Hollow, Castellan, and Dunmere. It also supplies irrigation water to farms in the lower valley.

The plan sets drought response rules, an infrastructure program, and a governance framework. It replaces the 2018 plan, which expired in December 2025.

2. Current Conditions

Annual rainfall in the catchment was 912 millimeters in 2025, compared with a 30-year average of 1,040 millimeters. Inflows were the second lowest in the past two decades.

The reservoir averaged 71% of capacity during 2025. The lowest level was 43% of capacity, reached in September 2025, after a dry summer.

Sediment accumulation continues to reduce usable storage by roughly 0.6% per decade, and a bathymetric survey is scheduled for 2027.

3. Water Demand

Average residential use is 118 liters per person per day, above the target of 105 liters. Peak summer demand reached 142 million liters per day in July 2025.

Agriculture accounts for 31% of total withdrawals, residential customers for 52%, and commercial and industrial users for the remaining 17%.

Demand is projected to grow 0.8% per year as the population increases, partly offset by efficiency measures.

4. Drought Response Stages

Stage 1 (Watch) begins when the reservoir falls below 60% of capacity. The Authority asks customers for voluntary reductions of 5%.

Stage 2 (Warning) begins below 50% of capacity. Reductions of 15% become mandatory and outdoor watering is limited to two days per week.

Stage 3 (Emergency) begins below 40% of capacity. Reductions of 30% are required and all non-essential outdoor use is prohibited. Violations may result in fines.

5. Infrastructure Program

Spillway rehabilitation will cost $18.4 million and is scheduled for completion in 2028. A new pump station will cost $6.2 million, and a leak detection program will cost $4.1 million.

The leak detection program is expected to reduce system water losses from 14% to 8% of treated water by 2031.

The remaining $13.0 million of the $41.7 million ten-year capital plan is held as a contingency reserve.

6. Governance and Monitoring

The Board of Directors reviews progress quarterly. A public dashboard showing reservoir levels is updated weekly, and the plan will be formally reviewed every three years.

The Board approved annual water rate increases of 3.5% for 2026 through 2028 to fund operations and debt service.

Customers can submit comments on the plan through the Authority's website until the end of March 2026.